

Prior knowledge-informed multi-task dynamic learning for few-shot machinery fault diagnosis

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ABSTRACT

The lack of machine fault data renders few-shot fault diagnosis important in modern industrial applications. Small sample sizes are sufficient to train a good model if prior diagnosis knowledge is incorporated into the learning process. This study presents a prior knowledge-informed multi-task dynamic learning model for few-shot fault diagnosis, which consists of a main task for fault identification and an auxiliary task for prior knowledge learning. Our framework starts with a selection of signal indicators as the source of prior knowledge based on domain knowledge. After that, a dynamic penalty on knowledge inconsistency, which is adjustable based on the training requirements, is designed for the auxiliary prior knowledge learning task. The knowledge learned in the auxiliary task is then utilized by the main task through a dedicated shared network structure, and thus the auxiliary information reduces the demand for training data in the main task. The proposed framework is applied to two fault diagnosis cases with small sample sizes. We demonstrate that compared with state-of-the-art methods, the proposed solution achieves an identification accuracy of up to 0.9803 with only 20 fault training samples.

1. Introduction

Complex equipment, such as aero-engines, can suffer substantial damage in the event of failures. Fault diagnosis is a pivotal component of Prognostics Health Management (PHM), acting as an intermediary between machine monitoring data and its health status (Guo et al., 2024). In industrial environments, condition monitoring and fault diagnosis are essential for reducing operational and maintenance costs while enhancing the reliability of machinery.

Vibration analysis is among the most widely employed techniques for machine fault diagnosis (Zhou et al., 2023). Traditionally, engineers use wavelet transform and other methods to analyze vibration monitoring data, relying on expert knowledge to assess the machine's health condition (Perez Colo et al., 2023). However, these approaches are heavily dependent on manual expertise, posing challenges for rapid and accurate data analysis. In the past decade, significant advancements in machine learning have led to the extensive application of deep neural networks in developing intelligent diagnostic models (Qin et al., 2020; Qiu et al., 2024; Wang et al., 2024). These models are capable of

analyzing large volumes of monitoring data to detect latent fault information. Despite their capacity for swift and precise data processing, these intelligent models require substantial amounts of training data. The performance of these models is directly correlated with the richness of the training datasets, with more comprehensive data yielding better results. In industrial contexts, machine failures are infrequent, making it challenging to collect sufficient fault data directly. Moreover, conducting fault simulation experiments in laboratories to generate such data is prohibitively expensive. Consequently, the availability of fault data for training intelligent models is very limited. Thus, developing effective fault diagnosis models with limited fault data remains a critical and practical challenge.

Scholars have conducted numerous investigations into the field of few-shot intelligent fault diagnosis, which aims to develop diagnostic models utilizing limited training samples (Zhang, Han et al., 2024; Zhang et al., 2024). These studies can be categorized into three main approaches: data augmentation-based, algorithm optimization-based, and transfer learning-based methods (Zhang et al., 2021). The first approach aims to enhance generalization performance through data

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augmentation techniques. For instance, Zhang, Li et al. (2023) proposed a method based on deep generative adversarial networks to augment machine monitoring data, thereby improving the diagnostic model's generalization capability with a small sample size. In Zhang, Kong et al. (2024), conditional least squares generative adversarial networks are proposed to generate fault samples for model training. Similarly, other data augmentation models like Variational Autoencoder (Huang et al., 2025) are also applied to machine fault diagnosis with limited samples. The second approach focuses on extracting richer fault features from the available limited fault data. For instance, Wang et al. (2023) utilized a lightweight model based on inverted residual block and collaborative self-attention for few-shot fault diagnosis. In Ye et al. (2025), a residual convolutional fusion network is presented to extract bearing fault features from limited samples. In another study (Zhang, Chen et al., 2023), a similarity-based contrastive-learning approach was employed for machine fault diagnosis using a limited number of training samples. The third approach involves leveraging relevant datasets to transfer diagnostic knowledge and address few-shot diagnosis tasks. For example, Kim and Youn (2019) presented a transferable siamese network for fault diagnosis, which can extract fault features of small samples by diagnosis knowledge transferring. Liu et al. (2021) proposed a deep domain adaptation-based transfer learning framework for bearing fault diagnosis with a small training dataset in the target domain. In Dong et al. (2022), a new dynamic transfer learning model is used to solve the bearing fault diagnosis problem with small samples. However, it is noteworthy that data augmentation-based methods consume substantial computational resources, posing challenges for developers. Additionally, algorithm optimization-based approaches may struggle to identify optimal solutions within the complex hypothesis set due to the limited data. Furthermore, transfer learning-based methods require appropriate transfer objects, and there is a risk of negative transfer in some cases.

Recently, there has been a growing interest among scholars in the machine learning community in integrating prior knowledge into the learning process of models, a paradigm known as prior knowledge-informed machine learning (PK-ML) (von Rueden et al., 2021). As illustrated in Fig. 1, PK-ML differs from traditional data-driven machine learning by being guided not only by training data but also by prior knowledge constraints. This integration allows prior knowledge to restrict the hypothesis set of the model, thereby narrowing its search space (Karniadakis et al., 2021). Consequently, the model is anticipated to require less training data to identify an appropriate final hypothesis. Thus, PK-ML is regarded as an effective approach to reducing the amount of training data necessary for model development. In the domain of machine fault diagnosis, scholars have amassed substantial expertise, encompassing fault characteristics, fault mechanisms, diagnostic rules, and related areas. These knowledge is derived from meticulous fault mechanism analyses and extensive engineering experience, rendering it both reliable and stable when applied to machine fault diagnosis tasks (Guo et al., 2020). Consequently, integrating such domain knowledge into the learning process of intelligent diagnostic models holds significant promise (Chen et al., 2022; Zhang et al., 2022). For example, Liu et al. (2025) proposed a network model with knowledge-informed finite impulse response filtering kernels for fault diagnosis under small samples. Lu et al. (2025) applied a prior knowledge-embedding convolutional autoencoder for fault diagnosis under small samples. Thus, by leveraging domain knowledge, it is feasible to develop satisfactory diagnostic models even with limited fault data.

Multi-task learning, which facilitates the transfer of knowledge learned from one task to others, offers an effective approach for integrating prior knowledge into models (Shu & Huang, 2023; Vandenhende et al., 2022). In model development, multi-task learning typically involves a main task along with several related auxiliary tasks, wherein the developers' prior knowledge can be embedded. For instance, in multi-task learning-based facial landmark detection (Zhang et al., 2014),

Table 1

The used symbols and their corresponding meanings.

Symbol	Meaning	Symbol	Meaning
x	Data sample	y	Sample label
D	Formed dataset	N	Total samples number
M	Sample dimension	L	Total labels number
T	Task	p_k	Prior features
L_c	Classification loss	L_k	Knowledge inconsistency loss
f	Model operation	λ	Loss weight
c	Output of convolutional layer	W	Feature vector
k	Kernel	b	Bias
r	Output of pooling layer	μ	Window size
s	Stride	$lrelu$	Activation function
q	Output of fully connected layer	t	Epoch number

the main task is facial landmark detection, while the auxiliary tasks include gender recognition and facial pose recognition, etc. These auxiliary tasks, informed by expert prior knowledge, assist the model in learning facial features pertinent to facial landmark detection. Consequently, the performance of the main task is enhanced by the integration of prior knowledge through the auxiliary tasks.

The previous review has demonstrated that integrating prior knowledge into the learning process of diagnostic models can significantly reduce their reliance on fault data. Concurrently, multi-task learning has been identified as an effective strategy for incorporating such knowledge. Drawing on these insights, this paper proposes a prior knowledge-informed multi-task dynamic learning network for few-shot machine fault diagnosis. The proposed method includes a main task focused on fault identification and a dynamic auxiliary task dedicated to learning prior diagnostic knowledge. Informed by fault diagnosis domain knowledge, we recognize that feature indicators of vibration signals, such as kurtosis and skewness, can partially reflect a machine's health status. Therefore, we first select a series of signal feature indicators to form a prior feature set, whose efficacy has been validated in numerous studies (Chen et al., 2018; Zhang et al., 2022). In the auxiliary task, our network aims to learn these prior features from the input data by optimizing a dynamic knowledge inconsistency loss. This dynamic loss is adjustable based on the real-time training progress of both the main and auxiliary tasks. Given the shared network structure, the knowledge acquired by the auxiliary task is also available to the main task, thereby easing the main task's learning burden and reducing its need for extensive training data. Consequently, the main task is expected to achieve accurate fault identification with a limited amount of training data.

The contributions of this paper can be summarized in the following.

- (1) A multi-task dynamic learning network, which can discover prior domain knowledge from input, is defined to extract highly reliable features from small samples data.
- (2) A prior knowledge-informed diagnosis method is proposed based on multi-task dynamic learning to deal with the challenging few-shot fault diagnosis issue.
- (3) Extensive experiments based on two mechanical fault simulation test benches demonstrate the effectiveness of the proposed method. Our solution has a significant advantage over state-of-the-art methods in small samples scenarios.

The rest of this paper is organized as follows. Section 2 describes the proposed method in detail. Section 3 verifies the superiority of the proposed method with two mechanical fault simulation experiments. Section 4 concludes the whole paper.

2. Methodology

Firstly, for the sake of better clarity, the symbols used in this section and their corresponding meanings can be found in Table 1.

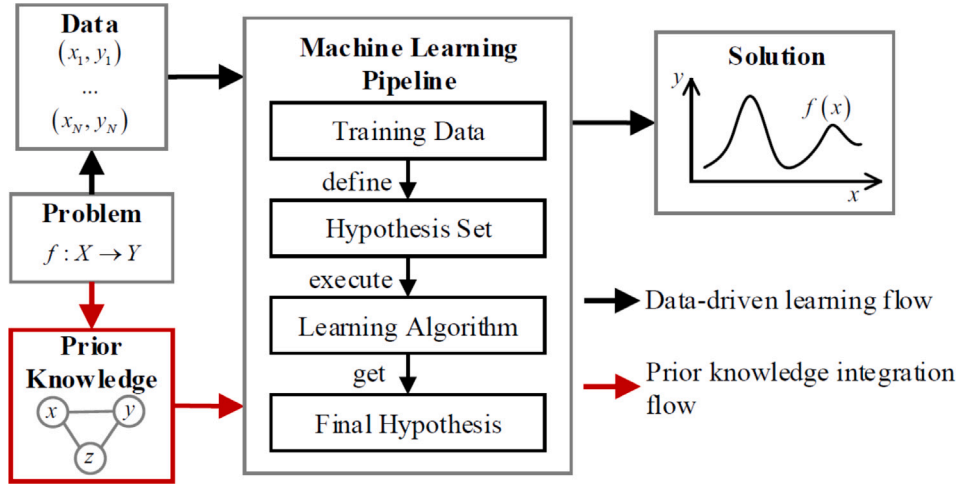


Fig. 1. Illustration of PK-ML.

2.1. Problem statement

This study concentrates on the condition monitoring and fault diagnosis of rotating machinery, specifically focusing on bearings and gears, which are essential components. The objective is to diagnose the fault location and severity of these monitored parts, providing crucial decision-making support for the prognostics health management of the machinery.

We analyze the machine's vibration signals, collected through vibration sensors and data acquisition instruments. The collected vibration signals from the bearings or gears are segmented and labeled to create a dataset $D = (x_i, y_i)_{i=1}^N$, where $x_i \in \mathbb{R}^{M \times 1}$ represents the i th sample with M vibration data points, and $y_i \in [L] \equiv \{1, 2, \dots, L\}$ is the health state label. Considering scenarios with a limited number of fault samples, we constrain N to be less than or equal to 20. Using this dataset, the study aims to develop a fault diagnosis method for rotating machinery that can learn the nonlinear mapping $f: \mathbb{R}^M \rightarrow [L]$ from vibration signals to health states, utilizing a small number of fault samples.

2.2. Overview of the proposed method

In industrial settings, the availability of machine fault data is typically quite limited. Existing intelligent diagnosis models, particularly those based on deep neural networks, require substantial amounts of fault data for training due to their numerous parameters. Recent research indicates that incorporating domain knowledge into the model training process can significantly reduce the need for extensive training data.

To address these challenges, this study introduces a multi-task dynamic learning network that leverages prior knowledge for machine fault diagnosis in scenarios with limited sample sizes. As illustrated in Fig. 2, the proposed method involves constructing a prior feature set of vibration signals informed by domain knowledge. The main task $T_1: x \rightarrow y$ focuses on fault diagnosis, while the auxiliary task $T_2: x \rightarrow p_k$ is dedicated to prior feature learning, where p_k represents the predefined prior feature set. The main task T_1 shares and utilizes the knowledge acquired by T_2 via a common feature extractor, thereby reducing the dependency on extensive training data. The training objectives of the proposed method are articulated as

$$f^* = \arg \min_{f \in F} \left(\lambda_c \sum_{i=1}^N L_c(f(x_i), y_i) + \lambda_k \sum_{i=1}^N L_k(f(x_i), p_{ki}) \right) \quad (1)$$

where $L_c(\cdot)$ is the classification loss in T_1 and λ_c is its weighting factor. $L_k(\cdot)$ is the knowledge inconsistency loss in T_2 and λ_k is its weighting factor.

Table 2

10 prior feature indicators.

Indicator	Equation	Indicator	Equation
Maximum	$p_1 = \max(x(m))$	Minimum	$p_2 = \min(x(m))$
Peak	$p_3 = \max x(m) $	Peak-Peak	$p_4 = \max(x(m)) - \min(x(m))$
Absolute mean	$p_5 = \frac{1}{M} \sum_{m=1}^M x(m) $	Square root amplitude	$p_6 = \left(\frac{1}{M} \sum_{m=1}^M \sqrt{ x(m) } \right)^2$
Variance	$p_7 = \frac{1}{M} \sum_{m=1}^M (x(m))^2$	Standard deviation	$p_8 = \sqrt{\frac{1}{M-1} \sum_{m=1}^M [x(m) - \bar{x}]^2}$
Root mean square	$p_9 = \sqrt{\frac{1}{M} \sum_{m=1}^M (x(m))^2}$	Kurtosis	$p_{10} = \frac{1}{M} \sum_{m=1}^M (x(m))^4$

Where $x(m)$ is a sample containing M data points, $\bar{x}$ is the mean value.

2.3. Prior feature set

In the domain of fault diagnosis, substantial diagnostic knowledge has been accumulated by experts, encompassing machine fault mechanisms, fault features, diagnostic rules, and feature extraction algorithms, all of which constitute the prior knowledge in this field (Guo et al., 2020). Notably, fault features include signal indicators such as kurtosis and skewness, which are indicative of a machine's health condition. These signal indicators are grounded in rigorous analyses of fault mechanisms and extensive expert experience, and their efficacy has been extensively validated in practical applications.

In this method, we select ten signal indicators to constitute the prior feature set $p_k = \{p_1, p_2, \dots, p_{10}\}$, where p_j is the j th prior indicator, as given in Table 2. We extract these ten prior features from the training data and normalize them using zero-mean normalization.

2.4. Feature extractor

In the proposed method, the main task T_1 and the auxiliary task T_2 share the feature extractor, which is the backbone network of the method. The feature extractor consists of convolutional layers, pooling layers, and fully connected layers, whose structure and parameter settings are shown in Table 3.

In the feature extractor, the output of the u th convolutional layer is c_u .

$$c_u^v = \text{lrelu} \left(\sum_{\tau \in W_u} r_{u-1}^v \times k_u^{\tau, v} + b_u^v \right) \quad (2)$$

where W_u is the feature vector in this layer, k_u and b_u are the kernel and bias of this layer. r_{u-1} represents the output of the $u-1$ th max-pooling

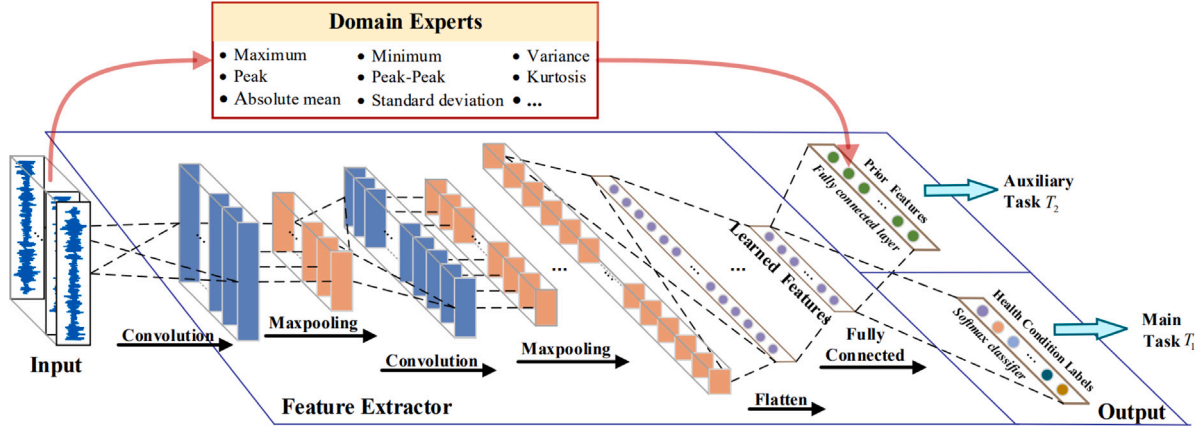


Fig. 2. Framework of the proposed method.

Table 3
The structure of the feature extractor.

Layer	Operation	Channels @ Kernel size* Stride/Pool size*Stride	Output shape	Activation function
Input	/	/	1*1024	/
Conv 1	1D Convolution	32@4*1	32*1024	lrelu
MP 1	Max pooling	2*2	32*512	/
Conv 2	1D Convolution	32@4*1	32*512	lrelu
MP 2	Max pooling	2*2	32*256	/
Conv 3	1D Convolution	64@4*1	64*256	lrelu
MP 3	Max pooling	2*2	64*128	/
Conv 4	1D Convolution	64@4*1	64*128	lrelu
MP 4	Max pooling	2*2	64*64	/
F	Flatten	/	4096	/
FC 1	Fully connected	/	1350	lrelu
FC 2	Fully connected	/	450	lrelu
FC 3	Fully connected	/	150	lrelu
FC 4	Fully connected	/	50	lrelu

layer.

$$r_{u-1}^o = \max_{a \in \mu} e^{a+0*s} \quad (3)$$

where μ and s denote the window size and the stride, respectively. $lrelu$ is the activation function

$$lrelu = \begin{cases} x, & x \geq 0 \\ 0.2 * x, & x < 0 \end{cases} \quad (4)$$

The output of the u th fully connected layer is q_u .

$$q_u = lrelu(q_{u-1}w_u + d_u) \quad (5)$$

where w_u and d_u are the weight and bias of this layer.

2.5. Dynamic learning strategy

As described in Eq. (1), the training objectives of the proposed method contains two items: classification loss L_c is designed for the purpose of identifying fault types, while the knowledge inconsistency loss L_k is devised to minimize the discrepancy between predicted features and prior features, thereby enabling the model to learn from prior knowledge.

2.5.1. Classification loss L_c

In T_1 , the learned features are fed to the Softmax classifier to be classified. The operation of the Softmax classifier can be described as

$$h(x_i) = \frac{1}{\sum_{l=1}^L e^{\theta_l^T x_i}} \begin{bmatrix} e^{\theta_1^T x_i} & e^{\theta_2^T x_i} & \dots & e^{\theta_L^T x_i} \end{bmatrix}^T \quad (6)$$

where θ is the parameter of the Softmax classifier.

The classification loss in T_1 is a cross-entropy loss, which can be written as

$$L_c = -\frac{1}{N} \sum_{i=1}^N \sum_{l=1}^L 1\{y_i = l\} \log(h(x_i)_l). \quad (7)$$

2.5.2. Knowledge inconsistency loss L_k

In T_2 , the learned features output a 10-dimensional vector $\tilde{p} = \{\tilde{p}_1, \tilde{p}_2, \dots, \tilde{p}_{10}\}$ by a fully connected layer to approximate the prior features $p_k = \{p_1, p_2, \dots, p_{10}\}$ by optimizing the Mean Square Error. Thus, the knowledge inconsistency loss L_k can be defined as

$$L_k = \frac{\sum_{i=1}^N \sum_{j=1}^{10} (p_j^i - \tilde{p}_j^i)^2}{10 * N}. \quad (8)$$

2.5.3. Dynamic weights assignment

In multi-task learning, the magnitudes of loss values may differ across tasks. If these losses are directly summed to derive the final model loss, one task may dominate the learning process (Vandenhende et al., 2022). Therefore, it is crucial to assign appropriate weights to each task's loss.

The objective of our model is to achieve optimal classification accuracy, with a focus on allocating weights to maximize this performance by dynamically adjusting the weight λ_k of the knowledge inconsistency loss L_k during training. Consequently, we directly set the weight λ_c to 1 and adjust the weight λ_k of the auxiliary task T_2 to optimize the performance of T_1 .

Through preliminary experiments, we observed that during the initial stages of training, L_k is typically smaller than L_c , and task T_2 converges faster, indicating that the model has a tendency to prioritize learning task T_2 . Therefore, to coordinate the training progress of the two tasks, we designed a dynamic weight assignment scheme for λ_k , as shown in Eq. (9). This scheme adjusts λ_k based on real-time loss values, preventing the model from being dominated by task T_2 during the early stages of training. Furthermore, it enables optimal convergence of the main task T_1 in the later stages of training, ultimately achieving the best fault classification accuracy for the model.

$$\lambda_k(t) = \begin{cases} \frac{L_c(t)}{L_k(t)}, & L_c(t) \geq L_k(t) \\ 1, & L_c(t) < L_k(t) \end{cases} \quad (9)$$

where t is the number of training epoch.

Finally, the training process of the proposed method can be summarized in Algorithm 1.

Algorithm 1 The training process of the proposed method

Input: The training set $D = (x_i, y_i)_{i=1}^N$, optimizer, training epoch E .

Initialize: The trainable network parameters.

- 1: Calculate the prior features set $p_k = \{p_1, p_2, \dots, p_{10}\}$ of the training dataset $D = (x_i, y_i)_{i=1}^N$ by Table 1;
- 2: **for** epoch = 1 to E **do**
- 3: Calculate the sample features by Eqs. (2), (3), and (5);
- 4: T_1 output the predicted health condition labels $h(x_i)$ by Eq. (6);
- 5: T_2 output the predicted prior features $\bar{p}$ by (5);
- 6: Calculate L_c and L_k by Eqs. (7) and (8);
- 7: Calculate the final loss by assigning weight $\lambda_k(t)$ using Eq. (9);
- 8: Update the network parameters using optimizer;
- 9: **end**

3. Experimental validation and discussion

3.1. Overview

In this section, two mechanical fault test experiments are conducted: the Spectra Quest (SQ) bearing fault test experiment and the Tian-Xian (TX) bearing and gear fault test experiment. The efficacy of the proposed method is verified and discussed using these two experimental datasets. Specifically, we validate the diagnostic performance of bearing under small sample scenarios using the SQ data and analyze the learning mechanism of the proposed method. Additionally, given that unlabeled data are readily available in industrial settings, we explore the potential of the proposed method to leverage unlabeled monitoring data using the TX data.

The validation process was carried out on a computer with a 64-bit Windows 10 operating system, featuring an Intel Core i3-4170 CPU @ 3.70 GHz. The programming language used was Python 3.6.12, with TensorFlow 1.13.1 as the running environment.

For all experiments, the optimizer employed for the proposed method was Adam, with a learning rate of 0.0005. Each training session comprised 100 epochs. Furthermore, each experiment was repeated ten times, with the average results used for analysis.

3.2. Methods for comparison

To verify the superiority of the proposed method, we selected six methods to conduct comparison experiments, including two methods related to the proposed method and four state-of-the-art diagnosis methods.

- PFS: We use the ten signal feature indicators in Table 2 as input, and the classification results are directly output by a Softmax classifier.
- DCNN: We use a deep convolutional neural network to process the data and output the health labels. The structure, hyper-parameters, and training settings of DCNN are the same as the main task T_1 .
- KEAE: Chen et al. (2024) proposed a knowledge embedded auto-encoder network for fault diagnosis under small samples, which also uses domain knowledge to reduce the amount of training data.
- R-Net: Yang et al. (2021) built a residual convolutional neural network with wide convolutional kernels and applied it to machine fault diagnosis under limited fault samples.
- S-Net: A multi-state Siamese neural network with a few-shot learning ability was presented in Wen, Xue et al. (2024) for machine fault diagnosis in the case of small samples.

- FCGAN: Zhang, Li et al. (2023) used a signal-augmented model based on generative adversarial networks, which can improve the generalization ability of the diagnosis model under small samples condition.
- PKCA: Lu et al. (2025) proposed a prior knowledge-embedding convolutional autoencoder for fault diagnosis of bearings under small samples.

3.3. Evaluation criteria

To more comprehensively evaluate the effectiveness of the proposed method, we have adopted three common evaluation criteria. The Accuracy criterion reflects the proportion of correctly predicted samples by the model to the total number of samples, and its formula is denoted as

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (10)$$

where TP represents the number of samples that are predicted to be positive and are actually positive, FP stands for the number of samples that are predicted to be positive but are actually negative, FN denotes the number of samples that are predicted to be negative but are actually positive, and TN signifies the number of samples that are predicted to be negative and are actually negative.

The F1-Score criterion comprehensively considers the proportion of fault samples that are correctly predicted, with its formula expressed as

$$F1 - Score = \frac{2TP}{2TP + FP + FN} \quad (11)$$

The Area Under the Curve (AUC) score assesses the classification performance of the model from a probabilistic perspective, where a higher AUC score indicates more accurate output probabilities from the model. Its formula is denoted as

$$AUC = \frac{\sum_{i \in \text{positive}} rank_i - M * (M + 1) / 2}{M * N} \quad (12)$$

where M and N represent the number of positive samples and negative samples, respectively, while $rank_i$ denotes the ranking index of the i th positive sample.

3.4. Case study based on SQ fault experimental data

3.4.1. Experiments and data description

The SQ mechanical fault test experiment was carried out on the SQ test bench. As shown in Fig. 3(a), the SQ test bench contains an induction motor, a rotor, a magnetic powder brake, a data recorder, and a vibration sensor. According to Chen et al. (2016), about 40% of motor faults are related to bearings. Thus, we chose to test the drive-end bearing of the motor, which is an NSK6203 bearing. A vibration acceleration sensor with a sensitivity of 50 mV/g was used to collect vibration signals. In the experiment, the rotating speed of the motor was set to 40 Hz, and the sampling frequency of the data recorder was 25.6 kHz.

As shown in Fig. 3(b), six single-point faults were machined on the inner race and the outer race of bearings using a micro air grinder to simulate different bearing faults. The six kinds of faults are minor inner race fault (IF-1), medium inner race fault (IF-2), severe inner race fault (IF-3), minor outer race fault (OF-1), medium outer race fault (OF-2), and severe outer race fault (OF-3). Besides, a bearing in normal condition (NC-0) was also involved in the test process. Therefore, the vibration signals of seven different health conditions were collected in the SQ fault test experiment. We selected 1024 consecutive data points as one data sample. The signals of each health condition were divided into 1125 data samples, which are the data sources for model training and testing.

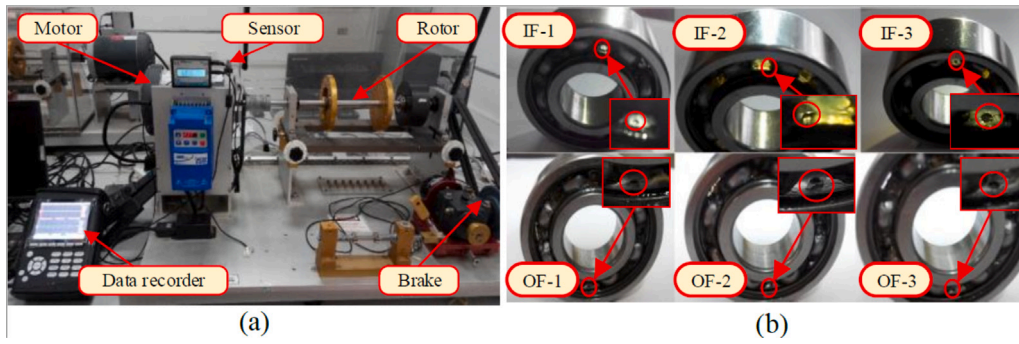


Fig. 3. SQ bearing fault experiment bench. (a) the SQ test bench, and (b) the faulty bearings.

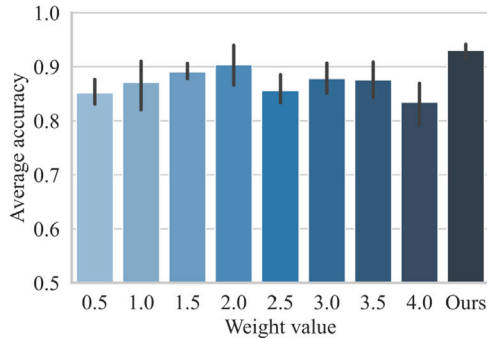


Fig. 4. Classification accuracy results using different weight values.

3.4.2. Effectiveness of the dynamic weight

First, we want to demonstrate the effectiveness of the dynamic weight $\lambda_k(t)$ in the proposed method. Since fixed weight is a common choice for multi-task learning, we selected eight fixed values from 0.5 to 4.0 as the weight for T_2 . The setting of this range was based on pre-experiments. The proposed method can achieve relatively high accuracy in this range. The experiments selected 10 data samples from each class as training samples and the rest as testing samples. The fault classification results of the method can be seen in Fig. 4.

When we use fixed weights, the method can achieve the highest accuracy of 0.9038 when the weight is 2. However, the accuracy increase to 0.9300 when the dynamic weight is applied. Thus, from the perspective of accuracy, the dynamic weight performs better. Further, we recorded the losses and the values of λ_k during the training process, and the results are shown in Fig. 5.

In Fig. 5(a), when we use fixed weight, there is a significant difference in the convergence speed between T_1 and T_2 , where the auxiliary task T_2 converges faster than the main task T_1 . This difference may cause the model to be more biased towards T_2 and not get the best diagnosis performance. The reason for this phenomenon is that the two tasks have different learning difficulties, and fixed weights cannot balance the real-time training progress. In Fig. 5(b), dynamic weight can effectively balance the learning speed of the two tasks and make the two tasks converge at a similar time. As a result, the proposed method can make the best use of the knowledge learned in T_2 and finally achieve the highest diagnostic accuracy in T_1 .

In Fig. 6, it illustrates the variations in the model's training loss and classification accuracy as the training progresses with 10 training samples. It can be observed that as the training loss gradually converges, the model's classification accuracy also gradually reaches its peak and remains stable.

3.4.3. Fault diagnosis under small samples

The proposed method addresses the few-shot fault diagnosis problem in industrial scenarios. Drawing inspiration from Zhang et al.

(2020), we define the condition of small samples as having no more than 20 training samples per class. This threshold of 20 is considered small in comparison to the total number of samples per class (1125) in the SQ experimental dataset. In the experiment, we selected 1, 5, 10, 15, and 20 samples from each class as training samples, with the remaining samples used for testing. The classification accuracy results of the proposed method, along with those of the six comparison methods, are presented in Table 4. The classification F1-Scores are presented in Table 5. The classification AUC results are presented in Table 6.

The results presented in the above Tables can be summarized as follows.

- **Effectiveness of Prior Features:** Even with only one training sample, the PFS achieves an accuracy of 0.7061, significantly surpassing the four other comparison methods. This indicates the effectiveness of the prior features in few-shot fault identification. PFS mitigates parameter overfitting in small sample conditions by utilizing manually calculated signal features rather than those extracted by neural networks.
- **Effectiveness of Multi-Task Learning:** The proposed method consistently outperforms the DCNN across all tasks. Given that DCNN shares the same structure and parameters as the main task in the proposed method, these results suggest that the multi-task learning framework enhances performance under small sample conditions.
- **Superiority Over State-of-the-Art Methods:** The proposed method demonstrates higher performance in all scenarios compared to the four state-of-the-art methods. This suggests that the proposed method may be more proficient in addressing few-shot fault diagnosis challenges in real industrial contexts. Additionally, as observed from the results presented in the three Tables, the prediction outcomes of the proposed method generally exhibit smaller standard deviations, suggesting that the proposed method possesses enhanced robustness under conditions of limited training samples.

Furthermore, we utilize t-SNE technique to reduce the dimensionality of the fault features extracted by the proposed method to two dimensions and visualize them, with the results presented in Fig. 7. The fault feature extraction illustrated in Fig. 7 is based on a condition with 10 training samples. It is evident that even under the condition of small training samples, the proposed method is still capable of extracting clearly distinguishable fault features from a large amount of bearing monitoring data.

3.4.4. Robustness analysis

In industrial settings, monitoring signals collected from machines often contain substantial background noise, which can obscure fault information and hinder accurate fault identification by diagnostic methods. Consequently, robustness to noise is a crucial factor in evaluating the practical applicability of diagnostic methods in industrial environments.

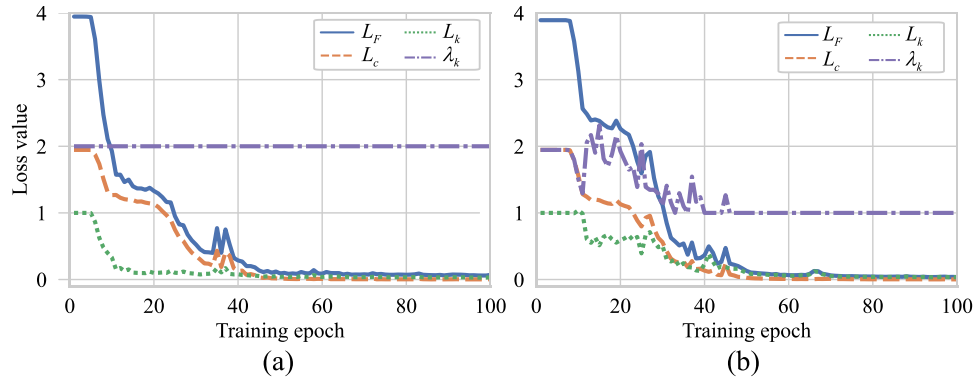


Fig. 5. Training losses with different λ_k . (a) $\lambda_k = 2$, and (b) λ_k is dynamic, L_F represents the final loss.

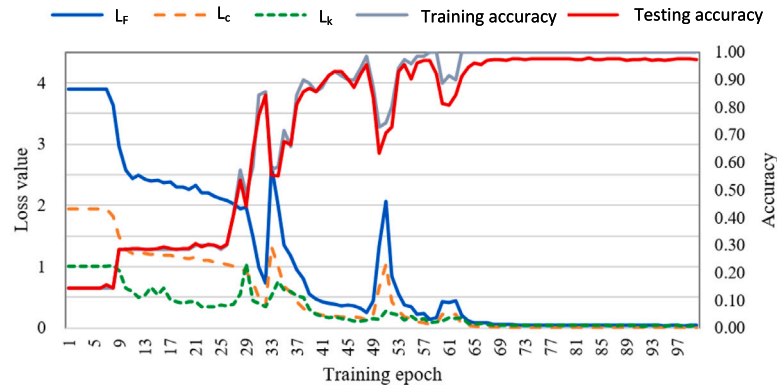


Fig. 6. Classification accuracy and loss value with the training progresses.

Table 4
Classification accuracy results based on SQ experimental data.

Model	Number of training samples				
	1	5	10	15	20
PFS	0.7061 $\pm$ 0.04	0.8087 $\pm$ 0.02	0.8265 $\pm$ 0.01	0.8864 $\pm$ 0.01	0.9069 $\pm$ 0.01
DCNN	0.1965 $\pm$ 0.03	0.6180 $\pm$ 0.11	0.8121 $\pm$ 0.05	0.8622 $\pm$ 0.02	0.9304 $\pm$ 0.02
KEAE	0.3167 $\pm$ 0.06	0.6056 $\pm$ 0.05	0.6479 $\pm$ 0.02	0.7620 $\pm$ 0.02	0.8723 $\pm$ 0.04
R-Net	0.1417 $\pm$ 0.01	0.5762 $\pm$ 0.06	0.5808 $\pm$ 0.04	0.7484 $\pm$ 0.04	0.8637 $\pm$ 0.03
S-Net	/	0.6071 $\pm$ 0.03	0.6189 $\pm$ 0.02	0.7309 $\pm$ 0.03	0.8339 $\pm$ 0.02
FCGAN	0.4370 $\pm$ 0.07	0.7438 $\pm$ 0.04	0.8165 $\pm$ 0.06	0.9010 $\pm$ 0.04	0.9375 $\pm$ 0.04
PKCA	0.6754 $\pm$ 0.05	0.7267 $\pm$ 0.04	0.8054 $\pm$ 0.04	0.8802 $\pm$ 0.03	0.9165 $\pm$ 0.02
Proposed	0.7322 $\pm$ 0.02	0.8333 $\pm$ 0.04	0.9297 $\pm$ 0.04	0.9676 $\pm$ 0.01	0.9803 $\pm$ 0.01

Table 5
Classification F1-Score results based on SQ experimental data.

Model	Number of training samples				
	1	5	10	15	20
PFS	0.6934 $\pm$ 0.03	0.7986 $\pm$ 0.02	0.8233 $\pm$ 0.01	0.8803 $\pm$ 0.01	0.8972 $\pm$ 0.01
DCNN	0.1732 $\pm$ 0.04	0.6312 $\pm$ 0.05	0.8056 $\pm$ 0.04	0.8676 $\pm$ 0.02	0.9281 $\pm$ 0.02
KEAE	0.3036 $\pm$ 0.05	0.6371 $\pm$ 0.04	0.6623 $\pm$ 0.02	0.7646 $\pm$ 0.02	0.8826 $\pm$ 0.02
R-Net	0.1557 $\pm$ 0.03	0.5361 $\pm$ 0.06	0.5834 $\pm$ 0.04	0.7512 $\pm$ 0.04	0.8701 $\pm$ 0.02
S-Net	/	0.6177 $\pm$ 0.03	0.6265 $\pm$ 0.02	0.7375 $\pm$ 0.03	0.8402 $\pm$ 0.02
FCGAN	0.4254 $\pm$ 0.05	0.7587 $\pm$ 0.04	0.8342 $\pm$ 0.06	0.9136 $\pm$ 0.04	0.9423 $\pm$ 0.03
PKCA	0.6682 $\pm$ 0.05	0.7184 $\pm$ 0.03	0.8133 $\pm$ 0.03	0.8783 $\pm$ 0.03	0.9097 $\pm$ 0.02
Proposed	0.7488 $\pm$ 0.02	0.8378 $\pm$ 0.03	0.9385 $\pm$ 0.04	0.9694 $\pm$ 0.01	0.9805 $\pm$ 0.01

To simulate noise-polluted monitoring signals typical of real-world scenarios, we artificially added varying levels of Gaussian noise to the monitoring signals. The signal-to-noise ratio (SNR) was used to quantify the noise intensity, with lower SNR values indicating higher noise intensity. In the experiment, we adjusted the SNR of the monitoring

signals from 20 dB to 0 dB incrementally. For illustrative purposes, the monitoring signals for the OF-1 fault class under different SNR conditions are shown in Fig. 8. Additionally, the number of training samples was set to 20. The diagnostic results of the proposed method compared with other methods are presented in Fig. 9.

Table 6
Classification AUC results based on SQ experimental data.

Model	Number of training samples				
	1	5	10	15	20
PFS	0.7123 $\pm$ 0.05	0.8145 $\pm$ 0.03	0.8312 $\pm$ 0.02	0.8921 $\pm$ 0.01	0.9178 $\pm$ 0.01
DCNN	0.1965 $\pm$ 0.03	0.6032 $\pm$ 0.4	0.8019 $\pm$ 0.05	0.8518 $\pm$ 0.02	0.9196 $\pm$ 0.02
KEAE	0.3024 $\pm$ 0.05	0.6147 $\pm$ 0.05	0.6328 $\pm$ 0.03	0.7516 $\pm$ 0.02	0.8612 $\pm$ 0.03
R-Net	0.1456 $\pm$ 0.02	0.5821 $\pm$ 0.05	0.5792 $\pm$ 0.04	0.7396 $\pm$ 0.04	0.8528 $\pm$ 0.03
S-Net	/	0.6124 $\pm$ 0.03	0.6087 $\pm$ 0.03	0.7215 $\pm$ 0.03	0.8247 $\pm$ 0.01
FCGAN	0.4218 $\pm$ 0.05	0.7326 $\pm$ 0.05	0.8054 $\pm$ 0.06	0.8924 $\pm$ 0.04	0.9223 $\pm$ 0.03
PKCA	0.6643 $\pm$ 0.04	0.7178 $\pm$ 0.04	0.7946 $\pm$ 0.03	0.8714 $\pm$ 0.03	0.9056 $\pm$ 0.02
Proposed	0.7187 $\pm$ 0.02	0.8245 $\pm$ 0.03	0.9186 $\pm$ 0.03	0.9587 $\pm$ 0.01	0.9692 $\pm$ 0.01

Table 7
Analysis of the computational burden of the model.

Methods	PFS	DCNN	KEAE	R-Net	S-Net	FCGAN	PKCA	Proposed
Training time (s)	2.19	150.31	217.19	371.35	128.27	1243.47	188.21	154.68
Testing time (s)	33.15	13.17	9.16	7.71	10.68	10.64	9.87	12.06
Parameters	/	6.24×10^6	1.40×10^6	3.14×10^5	4.71×10^5	2.01×10^6	8.61×10^6	6.24×10^6
FLOPs	/	6.38×10^7	2.28×10^7	3.22×10^6	5.56×10^6	4.15×10^7	4.05×10^8	6.38×10^7

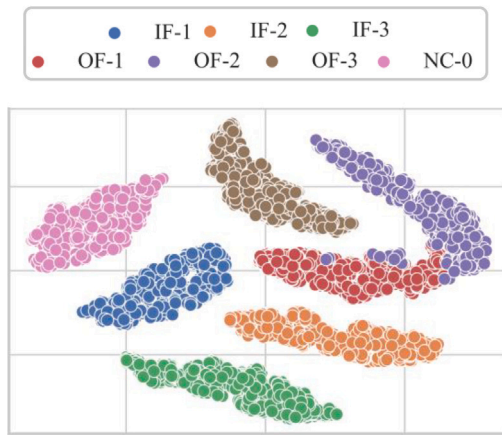


Fig. 7. Fault feature extraction of the proposed method using 10 training samples.

As depicted in Fig. 9, diagnostic accuracy decreases significantly with increasing noise intensity for all methods. Nevertheless, the proposed method consistently achieves the highest diagnostic accuracy across almost all tasks. Even at 0 dB, where the noise intensity matches that of the original signal, the proposed method maintains an accuracy of 0.7698. This demonstrates that, compared to other methods, the proposed method is more effective in handling noisy monitoring signals in industrial environments, thereby facilitating satisfactory engineering applications.

3.4.5. Computational burden

Given that computational resources are typically constrained in industrial settings, the requirement for computational resources has become a crucial metric for assessing diagnostic models. Consequently, Table 7 presents the training time, testing time, the number of parameters, and the floating point operations (FLOPs) for each of the seven methods. It is important to note that the results in Table 7 are derived based on ten training samples.

In Table 7, the parameter count for PFS is omitted as the features employed are derived through manual calculation rather than neural networks. Compared to the state-of-the-art methods (excluding PKCA), the proposed method exhibits slightly higher parameter count and computational complexity. Regarding training time, the FCGAN, which relies on data augmentation, demands a longer duration than the other methods. For testing time, although the proposed method is slower than the four state-of-the-art methods, its performance remains

acceptable for industrial scenarios. Specifically, the proposed method takes 12.06 s to process 7*1115 samples simultaneously, a duration that corresponds to a monitoring period of 312.2 s, thereby providing sufficient coverage.

3.4.6. Learning mechanism of the multi-task network

In this section, we elucidate the learning mechanism of the proposed method by examining the network weights. The neural network learning process involves adjusting these weights. The essence of our method is to leverage knowledge acquired from the auxiliary task to guide the learning process of the main task, meaning the weights of the main task are influenced by the auxiliary task's learned knowledge.

To observe and evaluate the impact of the multi-task structure, we record the weight distribution P_k of the auxiliary task and the weight distribution P_m of the entire proposed method. The distribution P_k is obtained when the method solely performs the auxiliary task, while the initial weight distribution is set to a normal distribution P_i . Given that the main and auxiliary tasks share the feature extractor's parameters, we plot the weight distribution of the feature extractor. The results are presented in Fig. 10. Additionally, to quantify the similarity between the weight distributions P_k and P_m , we calculate the Wasserstein Distance, a standard metric for assessing distribution similarity.

In Fig. 10, the weight distributions exhibit a notable change in the first convolutional layer compared to the initial distribution P_i . Specifically, the Wasserstein Distance between P_k and P_m in this layer is 0.003102, indicating a high degree of similarity between the two distributions. This similarity suggests that the weights of the proposed multi-task network are significantly influenced by the auxiliary task.

Furthermore, we explore the relationship among the distributions P_m , P_k , and P_c in the classification task, which is guided solely by training data. As illustrated in Fig. 11, the distribution P_c is derived when the proposed method performs only the main task. The distribution P_c exhibits a certain symmetry, with denser distributions on both sides and a sparser distribution in the middle. In contrast, P_k does not display a markedly lower middle distribution compared to the sides. Meanwhile, P_m integrates the characteristics of both P_k and P_c . This observation indicates that the weights of the multi-task learning network are guided by both the training data and the prior features in the auxiliary task.

3.5. Case study based on TX fault experimental data

3.5.1. Experiments and data description

The TX fault test experiment was carried out on the TX test bench. As shown in Fig. 12(a), the TX test bench includes a motor, a gearbox, a data recorder, and vibration acceleration sensors. Unlike the SQ

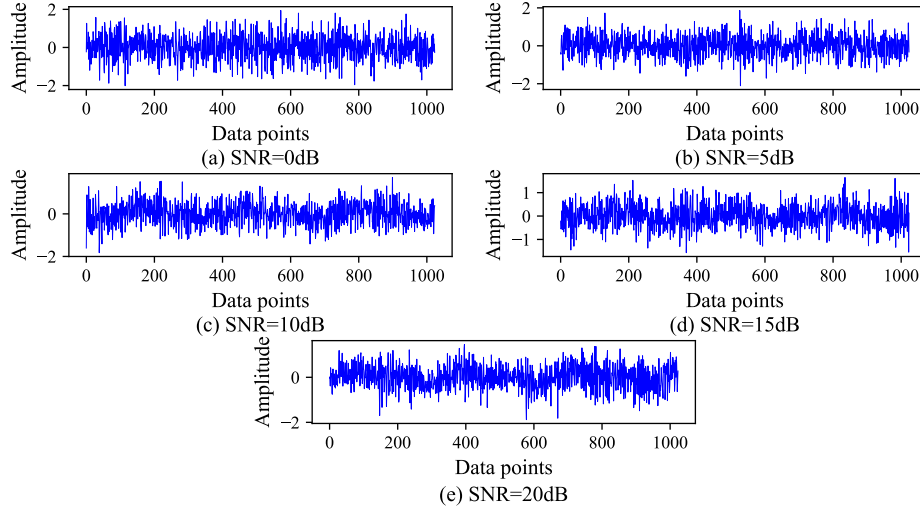


Fig. 8. Vibration signals of the OF-1 fault class under different SNRs.

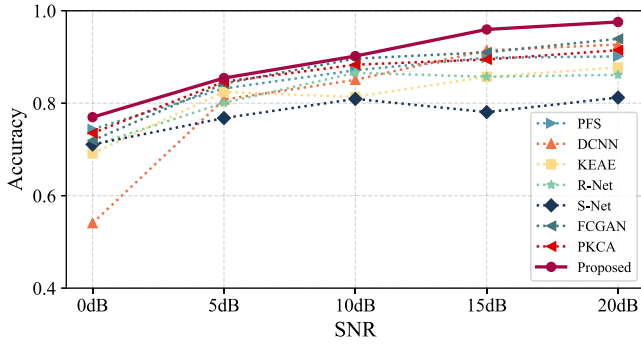


Fig. 9. Fault diagnosis accuracy results under different SNRs.

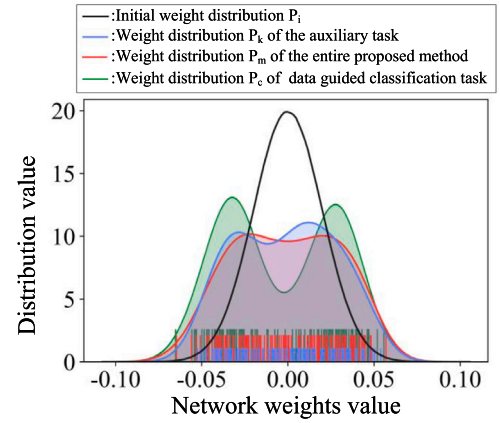


Fig. 11. The relationship between the three weight distributions of the feature extractor.

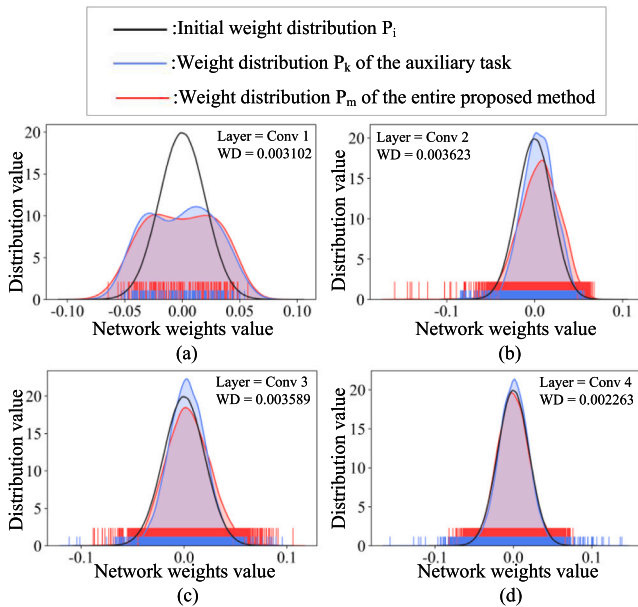


Fig. 10. The weight distributions in the feature extractor. WD represents the Wasserstein Distance. (a) Weight distributions in Conv 1 layer, (b) Weight distributions in Conv 2 layer, (c) Weight distributions in Conv 3 layer, (d) Weight distributions in Conv 4 layer.

experiment, the tested objects in the TX experiment include bearing and gear. The tested bearing and gear are installed in the gearbox, as shown in Fig. 12(b). In the experiments, we machined six single-point faults for each tested bearing and a fault on the tested gear. We conducted seven experiments to collect vibration data for these faulty objects. Besides, we also collected vibration data in normal conditions. The sensitivity of the used sensors is 50 mV/g, the speed of the motor was 800 rpm, and the sampling frequency of the data recorder was 5 kHz.

The eight health conditions can be abbreviated as follows: bearing cage fault (CF-0), bearing roller fault (RF-0), bearing outer race spot weld fault (OF-0), bearing outer race minor fault (OF-1), bearing outer race medium fault (OF-2), bearing outer race severe fault (OF-3), gear fault (GF-0), and normal condition (NC-0). We use 1024 consecutive data points as one data sample, and the data in each health condition are divided into 774 data samples.

3.5.2. Fault diagnosis under small samples

We conducted experiments based on the proposed method and the six related methods. The experimental results are shown in Tables 8, 9, and 10.

In the above Tables, the proposed method performs better than the six related methods. Besides, we also give the confusion matrix of the classification results and the feature visualization results of the method when using 20 training samples. The results are shown in Fig. 13. The

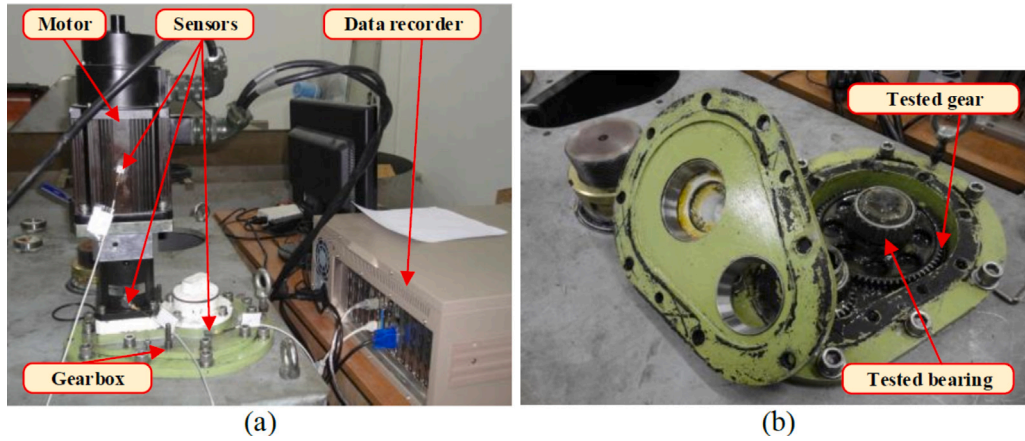


Fig. 12. TX bearing fault experiment bench. (a) TX test bench, and (b) tested gearbox with faulty bearings.

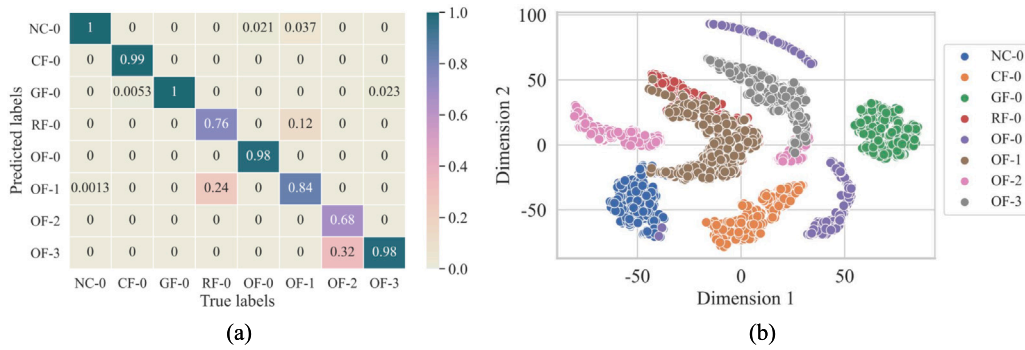


Fig. 13. Classification results with 20 training samples. (a) Classification confusion matrix, and (b) Extracted feature visualization.

Table 8

Classification accuracy results based on TX experimental data.

Model	Number of training samples				
	1	5	10	15	20
PFS	0.5099 ± 0.02	0.7028 ± 0.02	0.7557 ± 0.01	0.7914 ± 0.01	0.8256 ± 0.02
DCNN	0.3061 ± 0.03	0.6072 ± 0.04	0.7132 ± 0.03	0.7823 ± 0.02	0.8537 ± 0.03
KEAE	0.1875 ± 0.03	0.5568 ± 0.03	0.6943 ± 0.04	0.7073 ± 0.01	0.8055 ± 0.02
R-Net	0.2107 ± 0.10	0.6647 ± 0.06	0.7370 ± 0.06	0.7817 ± 0.06	0.8553 ± 0.04
S-Net	/	0.6788 ± 0.02	0.7531 ± 0.02	0.8256 ± 0.02	0.8639 ± 0.03
FCGAN	0.4647 ± 0.07	0.7544 ± 0.10	0.7654 ± 0.09	0.8325 ± 0.06	0.8442 ± 0.04
PKCA	0.4366 ± 0.08	0.7113 ± 0.08	0.7536 ± 0.04	0.8278 ± 0.04	0.8354 ± 0.03
Proposed	0.5101 ± 0.05	0.7695 ± 0.09	0.7801 ± 0.04	0.8462 ± 0.03	0.8806 ± 0.02

Table 9

Classification F1-Score results based on TX experimental data.

Model	Number of training samples				
	1	5	10	15	20
PFS	0.5213 ± 0.03	0.6914 ± 0.02	0.7469 ± 0.01	0.7786 ± 0.01	0.8128 ± 0.01
DCNN	0.3187 ± 0.03	0.5948 ± 0.06	0.7045 ± 0.03	0.7719 ± 0.02	0.8412 ± 0.03
KEAE	0.1894 ± 0.04	0.5476 ± 0.03	0.6859 ± 0.05	0.7127 ± 0.01	0.7943 ± 0.02
R-Net	0.2231 ± 0.06	0.6539 ± 0.05	0.7284 ± 0.04	0.7731 ± 0.04	0.8467 ± 0.03
S-Net	/	0.6692 ± 0.03	0.7423 ± 0.02	0.8148 ± 0.02	0.8527 ± 0.02
FCGAN	0.4532 ± 0.05	0.7456 ± 0.07	0.7568 ± 0.05	0.8231 ± 0.05	0.8338 ± 0.03
PKCA	0.4258 ± 0.06	0.7027 ± 0.05	0.7448 ± 0.04	0.8186 ± 0.04	0.8246 ± 0.02
Proposed	0.5249 ± 0.04	0.7587 ± 0.08	0.7716 ± 0.04	0.8374 ± 0.03	0.8798 ± 0.02

results show that the proposed method can distinguish different classes of vibration data using a small number of training samples even in the presence of multiple faulty objects.

3.5.3. Fault diagnosis with unlabeled data

In industrial scenarios, labeled monitoring data is rare because labeling data is a labor-intensive task. Meanwhile, unlabeled monitoring data is much easier to obtain. In fault diagnosis, many papers

Table 10
Classification AUC results based on TX experimental data.

Model	Number of training samples				
	1	5	10	15	20
PFS	0.4932 $\pm$ 0.02	0.6867 $\pm$ 0.02	0.7423 $\pm$ 0.01	0.7748 $\pm$ 0.01	0.8124 $\pm$ 0.01
DCNN	0.3205 $\pm$ 0.03	0.5916 $\pm$ 0.03	0.6998 $\pm$ 0.03	0.7657 $\pm$ 0.02	0.8372 $\pm$ 0.02
KEAE	0.1906 $\pm$ 0.03	0.5485 $\pm$ 0.03	0.6827 $\pm$ 0.03	0.7004 $\pm$ 0.01	0.7912 $\pm$ 0.02
R-Net	0.2253 $\pm$ 0.08	0.6563 $\pm$ 0.06	0.7294 $\pm$ 0.04	0.7732 $\pm$ 0.03	0.8477 $\pm$ 0.02
S-Net	/	0.6735 $\pm$ 0.02	0.7415 $\pm$ 0.02	0.8132 $\pm$ 0.02	0.8516 $\pm$ 0.01
FCGAN	0.4483 $\pm$ 0.05	0.7439 $\pm$ 0.07	0.7685 $\pm$ 0.06	0.8241 $\pm$ 0.06	0.8326 $\pm$ 0.05
PKCA	0.4202 $\pm$ 0.07	0.6998 $\pm$ 0.05	0.7427 $\pm$ 0.04	0.8194 $\pm$ 0.04	0.8271 $\pm$ 0.02
Proposed	0.5245 $\pm$ 0.03	0.7634 $\pm$ 0.05	0.7745 $\pm$ 0.03	0.8378 $\pm$ 0.03	0.8782 $\pm$ 0.02

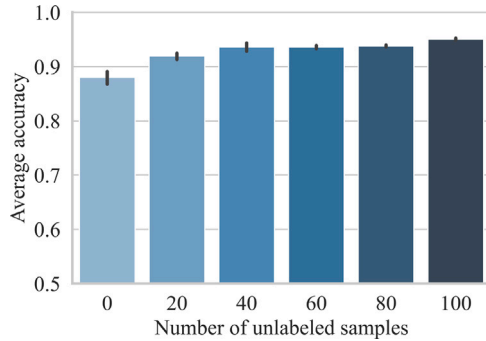


Fig. 14. Classification accuracy results with different unlabeled training samples.

focus on using unlabeled data to improve the performance of diagnosis models (Ran et al., 2024; Wen, Yang et al., 2024).

The proposed method also enables the use of unlabeled data. In the proposed method, the auxiliary task learns prior features from raw data, and this process does not require data labels. Therefore, we can dedicate additional unlabeled data to the training of the auxiliary task. In this case, the proposed method can learn prior knowledge from more data, which is expected to improve the proposed method's performance further.

To verify the enhancement effect of unlabeled data, we fixed the number of labeled training samples to 20. We took 20, 40, 60, 80, and 100 unlabeled samples to learn the auxiliary task. We recorded the fault identification accuracy in each case, and the results are shown in Fig. 14.

From Fig. 14, it can be found that the unlabeled samples helped the proposed method to obtain a better performance. When no unlabeled samples are involved in training, the proposed method's accuracy is 0.8806. The accuracy increases to 0.9516 when 100 unlabeled samples are used. The confusion matrix and the feature visualization results when the number of unlabeled samples is 100 are given in Fig. 15. Compared to Fig. 13, the use of unlabeled data allows the proposed method to obtain stronger classification ability and extract more discriminative features.

3.6. Discussion

The advantages of this method are summarized as follows. Firstly, the method enables accurate fault identification with a minimal amount of training data, which is particularly advantageous given the limited availability of fault monitoring data in industrial settings. Secondly, the method demonstrates robustness to noise, making it suitable for real industrial environments where strong noise pollution is prevalent. Thirdly, it can leverage unlabeled data to enhance diagnostic performance, thereby increasing its practical applicability in industrial contexts.

Compared with the state-of-the-art diagnostic methods, our approach demonstrates certain advantages in diagnostic accuracy, achieving a peak accuracy of 0.9803 with only 20 training samples. In

terms of computational complexity, we employ floating point operations (FLOPs) as the metric for measurement. Except for PKCA, the computational cost in FLOPs of the proposed method is slightly higher than that of several other selected state-of-the-art methods. Regarding the network backbone, our method employs a standard Convolutional Neural Network (CNN), which is relatively easier to design compared to the backbones used in other methods. In terms of knowledge utilization, our method, along with KEAE and PKCA, leverages domain knowledge in fault diagnosis. In terms of application scalability, we believe that other methods, with appropriate adjustments, have the potential to be applied to other diagnostic scenarios. For instance, FCGAN can be adjusted to address fault diagnosis in imbalanced datasets by modifying the data generation ratio. The proposed method exhibits superior application scalability due to its simple and easily replaceable network backbone and the universal applicability of the utilized knowledge in fault diagnosis. For example, the proposed method can select signal features more sensitive to gear faults as prior features, thereby enabling gear fault diagnosis with limited samples. Comprehensive comparison results are presented in Table 11.

Lastly, regarding future research directions, in the short term, we will focus on the theoretical analysis and optimization of the proposed method. This specifically encompasses the optimal design of dynamic weight assignment functions, enhancement of model stability, and control over overfitting and local minima. In the medium term, our research direction will be the application of the proposed method in bearing diagnosis across various scenarios, including unbalanced data scenarios and variable operating conditions. The long-term research directions encompass two aspects. Firstly, we aim to incorporate other fault diagnosis domain knowledge, such as fault mechanisms (Ha & Fink, 2023; Zhu et al., 2024). Secondly, we plan to extend the application of the proposed method to the diagnosis of other mechanical components, such as gearboxes and rotor systems. Detailed research plans can be found in Table 12.

4. Conclusion

As machine fault data are not always available, it is a challenge for building a data-driven diagnostic model with small samples. In this study, we propose a prior knowledge-informed multi-task dynamic learning-based method for few-shot fault diagnosis. This method achieves both prior diagnostic knowledge learning and faults identification. Due to the shared network structure, the method extracts highly reliable fault features with only a small amount of fault data. Two few-shot fault diagnosis scenarios are designed to verify the effectiveness of the proposed method in different task conditions. Extensive experimental results show that the proposed method can achieve more reliable fault identification with extremely limited fault samples compared to state-of-the-art methods. The proposed method achieves an diagnosis accuracy of up to 0.9803 with only 20 fault training samples. Moreover, the performance of the method can be further improved by using unlabeled data. Thus, the proposed method is a powerful tool to overcome the diagnostic challenges posed by the scarcity of fault samples.

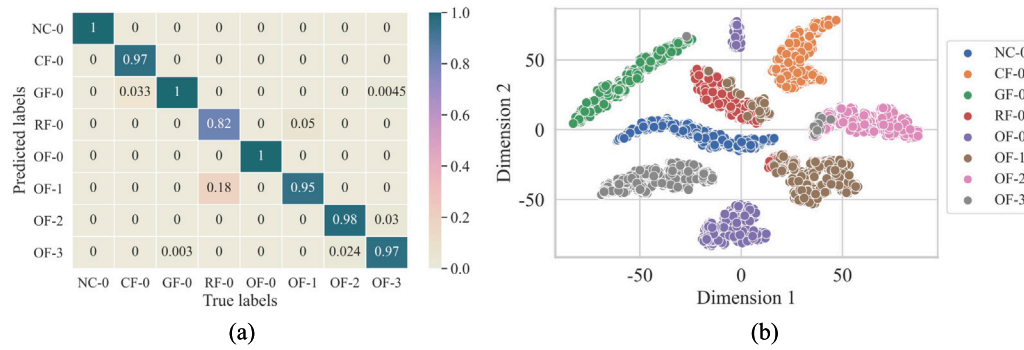


Fig. 15. Classification results with 20 labeled samples and 100 unlabeled samples. (a) Classification confusion matrix, and (b) Extracted feature visualization.

Table 11

Comparison of the proposed method with existing state-of-the-art methods.

Methods	Accuracy (20 training samples)	FLOPs	Network backbone	Knowledge utilization	Application scalability
KEAE	0.8723 ± 0.04	2.28×10^7	Multi-scale convolutional autoencoder	✓ (fault features)	✓
R-Net	0.8637 ± 0.03	3.22×10^6	Wide kernel residual CNN	✗	✓
S-Net	0.8339 ± 0.02	5.56×10^6	Siamese CNN	✗	✓
FCGAN	0.9375 ± 0.04	4.15×10^7	Regularized CNN	✗	✓
PKCA	0.9165 ± 0.02	4.05×10^8	Convolutional autoencoder	✓ (fault features)	✓
Proposed	0.9803 ± 0.01	6.38×10^7	Standard CNN	✓ (fault features)	✓ (better)

Table 12

Future research directions of the proposed method.

Research stage	Detailed research directions
Short-term directions	1. Optimal design of dynamic weight allocation functions 2. Enhancement of model training stability 3. Control over overfitting and local minima
Medium-term directions	1. Diagnostic applications in imbalanced data scenarios 2. Diagnostic applications under varying operating conditions
Long-term directions	1. Utilization of other diagnostic knowledge 2. Diagnostic applications to other mechanical components

CRedit authorship contribution statement

Tianci Zhang: Writing, Original draft, Methodology. **Jinglong Chen:** Review & Editing, Supervision. **Zhisheng Ye:** Writing, Review & Editing. **Wenting Liu:** Review & Editing. **Jinyuan Tang:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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